HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

8 December 2005

Time allowed 2 hours

Attempt all **7** questions

Total Marks 60

Only approved calculators may be used.

NUMERICAL SOLUTIONS

1. Fill in the missing values in the following table:

i	v	d	$i^{(3)}$	$d^{(2)}$
0.11	0.900901	0.099099	0.105733	0.101684

2. (a) £20.50
   (b) £26.49

3. (a) 38.9927
   (b) 8.6957
   (c) 316.545

4. (a) Bookwork
   (b) £161,772.44

5. (a) (i) £7,822.67
   (ii) £81,960.15
   (iii) £4,917.61
   (b) £12,191.80

6. (a) £91.19
   (b) 6.69% per annum

7. (a) MWRR ≈ 0.045.
   (b) The yield in 2004–5 is $i_1 = 0.06$. The yield in 2006–7 is $i_2 = 0.02836$. Hence LIRR is given as the solution of:

$$(1 + \text{LIRR})^4 = (1 + i_1)^2(1 + i_2)^2 \Rightarrow \text{LIRR} = 0.044060.$$

   (c) TWRR is given by:

$$\left(\frac{102}{100}.\frac{129}{122}.\frac{89.825}{79}\right)^{1/4} - 1 = 0.052325$$